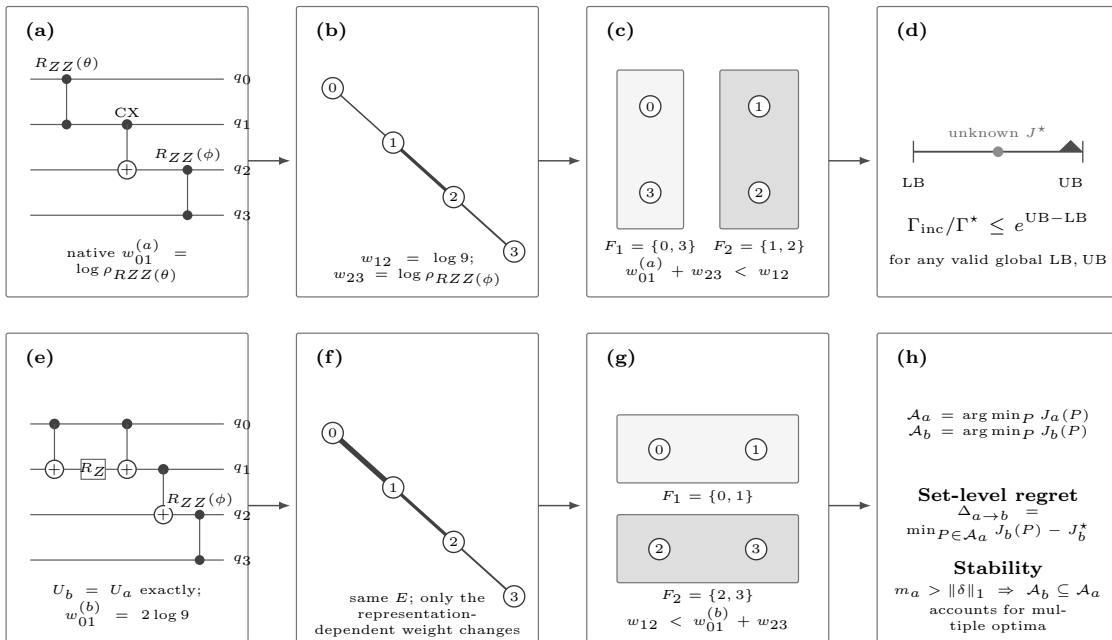


Semantics-preserving representation change and cut placement

Same logical unitary U and feasible placement space $\mathcal{F}$

$R_{ZZ}^{01}(\theta) = CX_{01} [I \otimes R_Z(\theta)] CX_{01}$; fixed K , capacities, logical wires, and fragment maps



Schematic four-qubit witness: the gate identity is exact; edge width denotes independent-QPD log cost.